Status	Finished
Started	Wednesday, 21 October 2026, 7:42 PM
Completed	Wednesday, 21 October 2026, 7:47 PM
Duration	4 mins 45 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Combining sorted sublists
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It does not require recursion
c. It sorts data using a single pass
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

this element

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Insertion sort

b. Bubble sort

c. Quick sort

d. Heap sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort

b. Insertion Sort

c. Merge Sort

d. Quick Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search

b. Binary Search

c. Bubble Sort

d. Merge Sort

The correct answer is: Merge Sort

this element

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Dividing the list into halves
- c. Bubbling up the largest element to its correct position
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

this element

Question **9**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Sorting

c. Merging

d. Searching

The correct answer is: Sorting

Question **10**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort

b. Quick sort

c. Selection sort

d. Merge sort

The correct answer is:
Merge sort

this element

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting decreases the efficiency of selection operations
c. Sorting makes it harder to search for items
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are split again into smaller sublists
c. They are discarded
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Bubble Sort

this element

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It only works with integer arrays
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently